Lattice QCD Analysis Methods

HMC Simulation

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1. Autocorrelation Function

For a sequence of measurements $\{O_0, O_1, O_2, \ldots, O_{N-1}\}$, the autocorrelation function is defined as:

$$\Gamma(t) = \frac{1}{N-t} \sum_{i=0}^{N-t-1} (O_i - \langle O \rangle)(O_{i+t} - \langle O \rangle) \tag{1}$$

where the average is

$$\langle O \rangle = \frac{1}{N} \sum_{i=0}^{N-1} O_i \tag{2}$$

2. Normalized Autocorrelation Function

The normalized autocorrelation function is:

$$\rho(t) = \frac{\Gamma(t)}{\Gamma(0)} \tag{3}$$

where $\Gamma(0)$ is the variance of the data:

$$\Gamma(0) = \mathrm{Var}(O) = \frac{1}{N} \sum_{i=0}^{N-1} (O_i - \langle O \rangle)^2 \tag{4}$$

3. Integrated Autocorrelation Time

The integrated autocorrelation time characterizes the number of effectively independent measurements:

$$\tau_{\mathrm{int}} = \frac{1}{2} + \sum_{t=1}^{W} \rho(t) \tag{5}$$

where W is the cutoff window. In our analysis, we use $W = N/2$.

4. Statistical Error

The error on the mean, accounting for autocorrelation, is:

$$\sigma_{\langle O \rangle} = \sqrt{\frac{2\tau_{\text{int}}}{N}}\sigma_O \tag{6}$$

where $\sigma_O = \sqrt{\Gamma(0)}$ is the standard deviation.

5. Plaquette Observable

The plaquette is the fundamental Wilson loop around a single plaquette:

$$P_{\mu\nu}(x) = \text{Tr}\left[U_\mu(x)U_\nu(x+\hat{\mu})U_\mu^\dagger(x+\hat{\nu})U_\nu^\dagger(x)\right] \tag{7}$$

The average plaquette:

$$\langle P \rangle = \frac{1}{N_t N_s^3 \cdot 6 \cdot 3}\sum_x \sum_{\mu<\nu} \text{Re}\left[P_{\mu\nu}(x)\right] \tag{8}$$

6. Topological Charge

The topological charge is computed using the clover field strength tensor:

$$Q = \frac{1}{32\pi^2}\sum_x \sum_{\mu\nu\rho\sigma} \varepsilon_{\mu\nu\rho\sigma}\text{Tr}\left[F_{\mu\nu}(x)F_{\rho\sigma}(x)\right] \tag{9}$$

The field strength tensor is constructed from 4 oriented plaquettes at each site:

$$C_{\mu\nu}(x) = \frac{1}{4}\left[P_{\mu\nu}(x) + P_{\nu,-\mu}(x) + P_{-\mu,-\nu}(x) + P_{-\nu,\mu}(x)\right] \tag{10}$$

$$F_{\mu\nu}(x) = \frac{1}{2}(C_{\mu\nu} - C_{\mu\nu}^\dagger)_{traceless} \tag{11}$$

7. Smearing

To improve the resolution of topological structures, we apply Stout smearing iteratively. For each smearing step, the links are updated as follows:

$$U_\mu(x) \to U'_\mu(x) = \frac{1}{\mathcal{N}}\left[U_\mu(x) + \alpha\,\Sigma_\mu(x)\right] \tag{12}$$

where $\Sigma_\mu(x)$ is the staple sum and $\mathcal{N}$ is a normalization factor ensuring U'_μ remains unitary after reunitarization via Gram-Schmidt.

The staple for direction μ is constructed from neighboring links:

$$\Sigma_\mu(x) = \sum_{\nu\neq\mu}\left[U_\nu(x)U_\mu(x+\hat{\nu})U_\nu^\dagger(x+\hat{\mu}) + U_\nu^\dagger(x-\hat{\nu})U_\mu(x-\hat{\nu})U_\nu(x-\hat{\nu}+\hat{\mu})\right] \tag{13}$$

After computing the updated link, we perform reunitarization to project back to SU(3).

Table 1: Smearing parameters used in this analysis

Parameter	Value	Description
α	0.1	Smearing strength
n_{iter}	8	Number of smearing steps

8. Simulation Parameters

Table 2: Lattice parameters and simulation details

Label	$N_T \times N_S^3$	β	Trajectories	Measurements
T16S8	16×8^3	5.9, 6.0, 6.5	5000	Gradient flow

Each trajectory consists of 10 molecular dynamics steps with $\tau = 1.0$.

9. Results: Topological Charge History (16×8^3)

The Monte Carlo history of the topological charge Q (measured at flow time $t = 2.0$) provides a direct visual indicator of autocorrelation severity. Long-lived plateaus in the history signal slow tunneling between topological sectors.

../../data_analyze/hmc/plots/T16_S8/beta5.9/topo_charge.png

Figure 1: Topological charge history and distribution for 16×8^3 at $\beta = 5.9$. The charge tunnels freely between integer sectors ($\sigma_Q = 0.95$), indicating good ergodicity.

../../data_analyze/hmc/plots/T16_S8/beta6.0/topo_charge.png

Figure 2: Topological charge history and distribution for 16×8^3 at $\beta = 6.0$. The history shows extended plateaus near integer values ($\sigma_Q = 0.63$), signaling the onset of critical slowing down.

../../data_analyze/hmc/plots/T16_S8/beta6.5/topo_charge.png

Figure 3: Topological charge history and distribution for 16×8^3 at $\beta = 6.5$. The charge is frozen near $Q = 0$ ($\sigma_Q = 0.06$), a clear sign of topological freezing in the deconfined phase.

10. Results: Topological Charge Autocorrelation (16×8^3)

The autocorrelation function $\rho(t)$ and integrated autocorrelation time τ_{int} for the topological charge Q quantify the observations from the MC histories above.

../../data_analyze/hmc/plots/T16_S8/beta5.9/autocorr_Q.png

Figure 4: Topological charge autocorrelation for 16×8^3 at $\beta = 5.9$: $\tau_{\text{int}} \approx 16.0$ MD time units.

../../data_analyze/hmc/plots/T16_S8/beta6.0/autocorr_Q.png

Figure 5: Topological charge autocorrelation for 16×8^3 at $\beta = 6.0$: $\tau_{\text{int}} \approx 38.1$ MD time units. The long autocorrelation time is consistent with the extended plateaus in the MC history.

../../data_analyze/hmc/plots/T16_S8/beta6.5/autocorr_Q.png

Figure 6: Topological charge autocorrelation for 16×8^3 at $\beta = 6.5$: $\tau_{\text{int}} \approx 1.2$ MD time units. The near-instant decorrelation reflects the frozen $Q \approx 0$ sector rather than good sampling.

11. Results: Topological Charge Distribution (16×8^3, Gradient Flow)

The topological charge distribution is measured at various gradient flow times t_{flow} for the 16×8^3 lattice. As flow time increases, the distribution sharpens toward integer values, demonstrating the smoothing effect of gradient flow on UV fluctuations.

../../data_analyze/hmc/plots/T16_S8/beta5.9/topo_distribution.png

Figure 7: Topological charge distribution at various flow times for 16×8^3 lattice at $\beta = 5.9$. The distribution develops peaks near integer Q values at larger t_{flow}.

../../data_analyze/hmc/plots/T16_S8/beta6.0/topo_distribution.png

Figure 8: Topological charge distribution at various flow times for 16×8^3 lattice at $\beta = 6.0$. The narrowing with flow time is more pronounced compared to $\beta = 5.9$.

../../data_analyze/hmc/plots/T16_S8/beta6.5/topo_distribution.png

Figure 9: Topological charge distribution at various flow times for 16×8^3 lattice at $\beta = 6.5$. At large flow times, Q is strongly concentrated near 0, consistent with topological freezing in the deconfined phase.

12. Discussion

The results show a striking observation: $\tau_{\text{int}}(Q) < \tau_{\text{int}}(P)$ for all parameter sets. This is opposite to the expected QCD behavior where topological charge typically has a longer autocorrelation time than the plaquette.

Possible explanations:

- The topological charge is frozen in the trivial sector ($Q \approx 0$)
- The lattice volumes are too small to observe topological transitions
- The Wilson flow scale w_0 may not be properly set

At $\beta = 6.3$, the plaquette autocorrelation time decreases significantly, indicating the system is in the deconfined phase.

Near the phase transition, the integrated autocorrelation time diverges:

$$\tau_{\text{int}} \propto \xi^z \sim |\beta - \beta_c|^{-z} \tag{14}$$

where ξ is the correlation length and $z \approx 2$ for local algorithms.

13. Fractional Topological Charge in $SU(N)/\mathbb{Z}_N$ Gauge Theory

Pure $SU(N)$ gauge theory possesses a $\mathbb{Z}_N$ 1-form (center) symmetry. Gauging this symmetry introduces a $\mathbb{Z}_N$ 2-form gauge field $B_{\mu\nu}$ on the lattice, which renders the topological charge fractional:

$$Q \;=\; \underbrace{-\frac{\varepsilon_{\mu\nu\rho\sigma}\, B_{\mu\nu} B_{\rho\sigma}}{8N}}_{\text{bundle label (fixed by } B_{\mu\nu})} \;+\; \underbrace{n}_{\text{instanton number (from } U)}\;, \qquad Q \in \frac{\mathbb{Z}}{N} \tag{15}$$

For $SU(2)/\mathbb{Z}_2$: Q takes values in half-integers $\{0, \pm\frac{1}{2}, \pm 1, \ldots\}$.

The key motivation is **topological freezing**: in standard HMC the instanton number n changes via tunneling over a barrier $\propto 1/a^2$ that diverges toward the continuum limit, while the bundle label $B_{\mu\nu}$ can be updated freely as a discrete variable.

13.1 $\mathbb{Z}_N$ 1-Form Symmetry

Definition

A **1-form symmetry** acts on line operators (Wilson loops) rather than point operators. Its conserved current is a 2-form $J_{\mu\nu}$, and the symmetry operator $U_\Sigma(k)$ is supported on a codimension-2 closed surface Σ. It acts on a Wilson loop $W_R(C)$ when the linking number $\mathrm{lk}(\Sigma, C) = 1$:

$$U_\Sigma(k)\, W_R(C) = e^{2\pi i k\, n_R/N}\, W_R(C) \tag{16}$$

where n_R is the N-ality of representation R.

Center of $SU(N)$

$Z(SU(N)) = \{\omega^k I_N : k = 0, \ldots, N-1\}$ with $\omega = e^{2\pi i/N}$, so $Z(SU(N)) \cong \mathbb{Z}_N$. The proof uses the irreducibility of the fundamental representation and Schur's lemma: any matrix commuting with all of $SU(N)$ must be λI_N with $\lambda^N = 1$.

Three Distinct Symmetries: from Global to Gauged

The $\mathbb{Z}_N$ 1-form symmetry follows the same logic as ordinary gauge theory: it begins as a **global** symmetry of the action, is then **promoted to local**, and a new gauge field B_p is introduced to restore invariance. To see why this works, we must first distinguish three transformations that act at different levels and leave the action invariant for *different reasons*.

(A) Standard $SU(N)$ gauge symmetry — acts on sites, conjugates links.
Parameter: $g(x) \in SU(N)$, one per site. Transformation:

$$U_\mu(x) \;\to\; g(x)\, U_\mu(x)\, g(x{+}\hat{\mu})^{-1} \tag{17}$$

This is a conjugation from both endpoints. The plaquette transforms as:

$$P_{\mu\nu}(x) \;\to\; g(x)\, P_{\mu\nu}(x)\, g(x)^{-1} \tag{18}$$

and the trace $\operatorname{tr} P$ is invariant because the trace is cyclic. This is the familiar lattice gauge invariance — it works for *any* $g(x)$, not just center elements. The $\mathbb{Z}_N$ center is just a subgroup; plugging $g(x) = z\,\mathbf{1}_N$ gives $z\,U_\mu\,z^{-1} = U_\mu$ — trivially nothing happens. This tells us nothing new.

(B) $\mathbb{Z}_N$ 1-form symmetry (global) — acts on links, left-multiplies.

This is a **completely different** transformation, *not* a special case of (A). It multiplies all links crossing a fixed codimension-1 hyperplane by a center element:

$$U_\mu(x) \;\to\; e^{2\pi i k/N}\,U_\mu(x) \qquad \text{(for all } x \text{ with } x_\mu = x_0\text{, fixed } \mu) \tag{19}$$

This is a *left-multiplication*, not a conjugation. It is not generated by any site-based gauge transformation — there is no $g(x)$ that produces a left-multiply without a compensating right-multiply from the next site.

Why is the action still invariant? A plaquette in the (μ,ν) plane at the hyperplane $x_\mu = x_0$ contains two $\hat{\mu}$-links: one going forward (picks up $e^{+2\pi i k/N}$) and one going backward (picks up $e^{-2\pi i k/N}$). These cancel. This is a **geometric** argument — the plaquette is a closed contractible loop, so it crosses the hyperplane an equal number of times in each direction. The cancellation has nothing to do with conjugation or trace cyclicity.

Key distinction:

	(A) $SU(N)$ gauge	(B) $\mathbb{Z}_N$ 1-form global
Parameter lives on	sites	links (hyperplane)
Acts on U_μ by	conjugation $g\,U\,g^{-1}$	left-multiply $z\,U$
Why $\operatorname{tr} P$ invariant	trace cyclicity	forward/backward cancellation
Applies to	any $SU(N)$ element	center elements $z \in \mathbb{Z}_N$ only

(C) $\mathbb{Z}_N$ 1-form symmetry (gauged) — local, requires B_p.

Promoting (B) to a *local* symmetry means allowing an independent $\lambda_\mu(x) \in \{0,\ldots,N-1\}$ on *every* link:

$$U_\mu(x) \;\to\; e^{2\pi i \lambda_\mu(x)/N}\,U_\mu(x) \tag{20}$$

Now the four links of a plaquette carry four independent phases, and their sum $(d\lambda)_p = \lambda_\mu(x) + \lambda_\nu(x{+}\hat{\mu}) - \lambda_\mu(x{+}\hat{\nu}) - \lambda_\nu(x)$ need not vanish. The plaquette picks up a net phase $e^{2\pi i (d\lambda)_p/N}$ and the standard action $\operatorname{Re}\operatorname{tr} P$ is **no longer invariant**. This is resolved by introducing B_p (described in the gauging section below).

Why $F_{\mu\nu}$ has N-ality 0 and the Link has N-ality 1

Under the 1-form transformation (B), the link picks up one uncompensated phase — it carries N-ality $n_R = 1$. More precisely, $U_\mu(x)$ is a matrix in the **fundamental representation**: it parallel-transports a vector $v \in \mathbb{C}^N$ from site $x{+}\hat{\mu}$ to x, so it carries one fundamental index.

The plaquette $P_{\mu\nu}$, being a closed loop, has all phases cancel — effective N-ality 0. In the continuum limit $P_{\mu\nu} \approx e^{ia^2 F_{\mu\nu}}$, so $F_{\mu\nu}$ inherits N-ality 0. This can also be seen algebraically: $F_{\mu\nu} = F^a_{\mu\nu} T^a$ where T^a are acted on by conjugation $T^a \to z\,T^a\,z^{-1} = T^a$ (the adjoint action), which is trivial for center elements.

Summary:

Object	Under 1-form (B)	Phase	N-ality
Link U_μ	$U_\mu \to z\, U_\mu$ (left-multiply)	$e^{2\pi i k/N}$	1
Plaquette $P_{\mu\nu}$ (closed loop)	forward/backward cancel	1	0
$F_{\mu\nu} = F^a T^a$ (continuum limit)	$T^a \to z\, T^a\, z^{-1}$	1	0
Polyakov loop (non-contractible)	crosses hyperplane once, net	$e^{2\pi i k/N}$	1

N-ality

The N-ality of an irreducible representation R with Young diagram $(\lambda_1 \geq \lambda_2 \geq \cdots)$ is $n_R = \sum_i \lambda_i \pmod N$. Under a center element: $\rho_R(\omega^k I_N) = e^{2\pi i k n_R/N} \cdot \mathbf{1}_{d_R}$.

In the fractional-Q papers, N-ality matters because the lattice action uses the **fundamental plaquette** ($n_R = 1$), so the B-field twist factor $e^{-2\pi i B/N}$ is non-trivial. An adjoint plaquette ($n_R = 0$) would decouple from B entirely.

From N-ality to Gauging: How the $\mathbb{Z}_N$ 2-Form Field Enters

As shown in (C) above, once the 1-form symmetry is made local, the plaquette picks up a net phase $(d\lambda)_p$ and the action breaks. To restore invariance, introduce a $\mathbb{Z}_N$-valued 2-form gauge field $B_p \in \{0, \ldots, N-1\}$ on each plaquette, transforming as:

$$B_p \;\to\; B_p + (d\lambda)_p \pmod N \tag{21}$$

The *gauged* action is:

$$S[U, B] = \beta \sum_{x,\, \mu<\nu} \frac{1}{N} \operatorname{Re}\operatorname{tr}\big[e^{-2\pi i B_p/N}\, P_{\mu\nu}(x)\big] \tag{22}$$

Under a $\mathbb{Z}_N$ gauge transformation, the plaquette picks up $e^{+2\pi i (d\lambda)_p/N}$ while the B factor shifts to $e^{-2\pi i (B_p + (d\lambda)_p)/N}$. The two phases cancel exactly:

$$e^{-2\pi i (B_p + (d\lambda)_p)/N} \cdot e^{+2\pi i (d\lambda)_p/N}\, P_p \;=\; e^{-2\pi i B_p/N}\, P_p \tag{23}$$

So the combination $e^{-2\pi i B_p/N} P_p$ is **gauge-invariant**.

This is the discrete analog of $U(1)$ minimal coupling: $\partial_\mu \to D_\mu = \partial_\mu - iA_\mu$ compensates for local phase rotations, just as B_p compensates for local $\mathbb{Z}_N$ rotations of the link. The hierarchy is shifted up by one form degree:

	$SU(N)$ gauge (0-form)	$\mathbb{Z}_N$ gauge (1-form)
Parameter lives on	sites: $g(x)$	links: $\lambda_\mu(x)$
Gauge field lives on	links: U_μ	plaquettes: B_p
Field strength on	plaquettes: $F_{\mu\nu}$	cubes: $(dB)_c$
"Matter" that transforms	quark $\psi(x)$	link $U_\mu(x)$

Why n_R is essential. The entire construction works because the link transforms with $n_R = 1$: it picks up a non-trivial phase $e^{2\pi i \lambda/N}$. If we had used an adjoint plaquette ($n_R = 0$), the plaquette would be invariant under *any* $\lambda_\mu(x)$ — local or global — so there would be nothing to gauge and no need for B_p. The B field would decouple completely.

Full Form of the Gauged Action

Combining all pieces, the lattice action coupled to the $\mathbb{Z}_N$ 2-form gauge field is:

$$S[U,B] = \beta \sum_{x,\,\mu<\nu} \frac{1}{N} \operatorname{Re}\operatorname{tr}\big[e^{-2\pi i B_{\mu\nu}(x)/N}\, P_{\mu\nu}(x) - \mathbf{1}\big] \tag{24}$$

where $P_{\mu\nu}(x) = U_\mu(x)\, U_\nu(x{+}\hat{\mu})\, U_\mu^\dagger(x{+}\hat{\nu})\, U_\nu^\dagger(x)$ is the standard plaquette built from periodic link variables, and $B_{\mu\nu}(x) \in \{0,\ldots,N{-}1\}$ is the $\mathbb{Z}_N$ 2-form field on each plaquette.

The phase factor $e^{-2\pi i B_{\mu\nu}(x)/N}$ is the minimal coupling that restores local $\mathbb{Z}_N$ gauge invariance. Setting $B_{\mu\nu}(x) = 0$ everywhere recovers the standard Wilson action. The $-\mathbf{1}$ ensures $S = 0$ for the classical vacuum $e^{-2\pi i B/N} P = \mathbf{1}$.

Under $\mathbb{Z}_N$ 1-form gauge transformation $\lambda_\mu(x)$:

$$U_\mu(x) \to e^{2\pi i \lambda_\mu(x)/N}\, U_\mu(x)\,, \qquad B_{\mu\nu}(x) \to B_{\mu\nu}(x) + (d\lambda)_{\mu\nu}(x) \pmod N \tag{25}$$

where $(d\lambda)_{\mu\nu}(x) = \lambda_\mu(x) + \lambda_\nu(x{+}\hat{\mu}) - \lambda_\mu(x{+}\hat{\nu}) - \lambda_\nu(x)$. The combination $e^{-2\pi i B/N} P$ is invariant, so $S[U,B]$ is gauge-invariant.

13.2 Continuum Origin of Fractional Q

Fractional topological charge requires non-trivial spacetime topology:

Spacetime	$H^2(M,\,\mathbb{Z}_N)$	Fractional Q?
$\mathbb{R}^4$	0 (contractible)	No, $Q \in \mathbb{Z}$
S^4	0	No
T^4	$\mathbb{Z}_N^6$	Yes, $Q \in \mathbb{Z}/N$
$\mathbb{R}^3 \times S^1$ (twisted b.c.)	$\mathbb{Z}_N$	Yes

On T^4, a gauge field need only be periodic up to a gauge transformation. Relaxing the cocycle condition by a center element gives an $SU(N)/\mathbb{Z}_N$ principal bundle, classified by 't Hooft fluxes $B_{\mu\nu} \in \mathbb{Z}_N$ (six integers, one per plane). Each non-trivial class forces $Q \in \mathbb{Z}/N$ by the Atiyah–Singer index theorem.

Fractional instantons are genuine BPS solutions ($F = \pm\tilde{F}$) with action $S = 8\pi^2/(Ng^2)$ and charge $Q = 1/N$.

13.3 't Hooft Flux and Twisted Boundary Conditions

This section traces how the 6 integers $B_{\mu\nu}$ enter the lattice action, starting from boundary conditions on T^4. The logic has four steps, and the key source of confusion is that B appears first as **6 constant integers** (from the cocycle), then as a **position-dependent field** $B_{\mu\nu}(x)$ (in the action), and finally is **gauge-fixed back** to 6 integers at the corner.

Step 1: Transition Functions on T^4

On a torus, the gauge field is periodic up to a gauge transformation. The transition function $\Omega_\mu \in SU(N)$ (one constant matrix per direction) specifies what happens when a link crosses the boundary in direction μ:

$$U_\nu(n + L\hat{\mu}) = \Omega_\mu\, U_\nu(n)\, \Omega_\mu^{-1} \tag{26}$$

Standard periodic boundary conditions are $\Omega_\mu = \mathbf{1}$.

Step 2: The Cocycle Gives 6 Constant Integers $B_{\mu\nu}$

At a corner of the torus where directions μ and ν meet, one can shift a link by $L\hat{\mu}$ first then $L\hat{\nu}$, or vice versa. Both must give the same result. Working out the two orderings:

$$\Omega_\mu\,\Omega_\nu = e^{2\pi i B_{\mu\nu}/N}\,\Omega_\nu\,\Omega_\mu \tag{27}$$

Since $\Omega_\mu, \Omega_\nu \in SU(N)$, the mismatch $(\Omega_\mu\Omega_\nu)^{-1}(\Omega_\nu\Omega_\mu)$ must commute with all of $SU(N)$. By Schur's lemma it is a center element $e^{2\pi i B_{\mu\nu}/N}\,\mathbf{1}$.

Here $B_{\mu\nu} \in \{0, \ldots, N{-}1\}$ is a **single constant integer** for each of the 6 planes (μ, ν) on T^4. It is the 't Hooft magnetic flux — a topological label that classifies which $SU(N)/\mathbb{Z}_N$ bundle the gauge field lives in.

$B_{\mu\nu} = 0$ for all planes corresponds to Ω_μ all commuting — this is the standard $SU(N)$ theory with periodic boundary conditions.

Step 3: Change of Variables — How $g_\mu(x)$ Produces $B_{\mu\nu}(x)$

The twisted links $\tilde{U}$ obey the boundary condition from Step 1:

$$\tilde{U}(x + L\hat{\mu},\, \nu) = g_\mu(x)^\dagger\, \tilde{U}(x, \nu)\, g_\mu(x + \hat{\nu}) \tag{28}$$

where $g_\mu(x)$ is the position-dependent transition function in direction μ. In Step 2 we used constant Ω_μ; in general $g_\mu(x)$ can vary along the boundary face, but its cocycle still satisfies $g_\mu(x)\, g_\nu(x{+}L\hat{\mu})\, g_\mu(x{+}L\hat{\nu})^\dagger\, g_\nu(x)^\dagger = e^{2\pi i B_{\mu\nu}(x)/N}\,\mathbf{1}$ at each boundary corner point x.

To do HMC, we want **strictly periodic** link variables $U(x{+}L\hat{\mu}, \nu) = U(x, \nu)$. The change of variables (Eq. (2.7) of arXiv:2501.00286) absorbs g_μ into the links:

$$\tilde{U}(x, \mu) = \begin{cases} U(x, \mu)\, g_\mu(x) & x_\mu = L{-}1 \\ U(x, \mu) & \text{otherwise} \end{cases} \tag{29}$$

The new U are strictly periodic. But when we substitute into the Wilson action, the boundary plaquettes that straddle $x_\mu = L{-}1$ now contain a factor of g_μ that does not cancel. This leftover is exactly the twist phase $e^{-2\pi i B_{\mu\nu}(x)/N}$:

$$\tilde{P}(x, \mu, \nu) = e^{-2\pi i B_{\mu\nu}(x)/N}\, P(x, \mu, \nu) \tag{30}$$

where $\tilde{P}$ is the plaquette in twisted variables and P is the plaquette in periodic variables. **This is how $B_{\mu\nu}(x)$ enters the action** — it is the $\mathbb{Z}_N$ phase left behind by the transition functions after absorbing them into periodic links.

The full action in periodic variables becomes:

$$S[U, B] = \beta \sum_{x,\, \mu<\nu} \frac{1}{N} \operatorname{Re}\operatorname{tr}\big[e^{-2\pi i B_{\mu\nu}(x)/N}\, P(x, \mu, \nu) - \mathbf{1}\big] \tag{31}$$

The relation between $g_\mu(x)$ and $B_{\mu\nu}(x)$ is:

- $g_\mu(x)$ is an $SU(N)$-valued function on the boundary — it contains much more information than B
- $B_{\mu\nu}(x)$ is the $\mathbb{Z}_N$ part of the cocycle of g_μ — it is the center phase that survives after the $SU(N)$ parts cancel in the action

- For constant $g_\mu = \Omega_\mu$: the cocycle gives a single constant $B_{\mu\nu}$, and only the boundary plaquettes are twisted
- For position-dependent $g_\mu(x)$: the cocycle gives $B_{\mu\nu}(x)$ that can vary along the boundary

Now $B_{\mu\nu}(x)$ is a position-dependent integer field. In principle it can be distributed across many plaquettes, as long as the total flux per plane equals $B_{\mu\nu}$: total flux per plane equals $B_{\mu\nu}$:

$$\sum_x B_{\mu\nu}(x) = B_{\mu\nu} \pmod N \tag{32}$$

The distribution of $B_{\mu\nu}(x)$ across plaquettes is $\mathbb{Z}_N$ gauge freedom — it is not physical.

Step 4: Gauge Fixing Back to 6 Integers (Corner Localization)

Using the $\mathbb{Z}_N$ 1-form gauge freedom $B_{\mu\nu}(x) \to B_{\mu\nu}(x) + \partial_\mu\lambda_\nu - \partial_\nu\lambda_\mu \pmod N$, one can "push" all the distributed flux into a single corner plaquette per plane. This is the gauge choice of Eq. (2.10) of arXiv:2501.00286:

$$B_{\mu\nu}(x) = \begin{cases} B_{\mu\nu} & x_\mu = L-1 \text{ and } x_\nu = L-1 \\ 0 & \text{otherwise} \end{cases} \tag{33}$$

Summary of the full chain:

$$\underbrace{B_{\mu\nu}}_{\text{6 constants from cocycle}} \xrightarrow{\text{change of variables}} \underbrace{B_{\mu\nu}(x)}_{\text{distributed on plaquettes}} \xrightarrow{\text{gauge fix}} \underbrace{B_{\mu\nu} \text{ at corner}}_{\text{back to 6 integers}}$$

The starting point and endpoint are both 6 integers — the position-dependent $B_{\mu\nu}(x)$ in the middle is just an intermediate representation that carries gauge redundancy.

Strict vs. Relaxed Cocycle

	Standard $SU(N)$	't Hooft flux $B_{\mu\nu} \neq 0$
Cocycle	$\Omega_\nu\Omega_\mu = \Omega_\mu\Omega_\nu$	Relaxed by center element
Bundle	$SU(N)$	$SU(N)/\mathbb{Z}_N$
Gluon A_μ well-defined?	Yes	Yes
Quark ψ well-defined?	Yes	No (picks up $e^{2\pi i/N}$)
Q	$\in \mathbb{Z}$	$\in \mathbb{Z}/N$
Sectors	1	N^6

The strict cocycle ($B = 0$) is not forced by gauge theory self-consistency — it is forced by requiring fundamental-representation quarks to be globally well-defined. In pure Yang–Mills, all N^6 bundle sectors are valid.

Why B Has No Kinetic Term

Unlike a continuous gauge field A_μ which has a kinetic term $F_{\mu\nu}^2$, the $\mathbb{Z}_N$ 2-form field $B_{\mu\nu}(x)$ has **no kinetic term** in the action. The full action is just the coupling to the plaquette — there is no additional $(dB)^2$ term. Two reasons:

1. B **is discrete** — it takes values in $\{0, \ldots, N-1\}$. One cannot define a continuous derivative of an integer-valued field, so a conventional kinetic term is not meaningful. B has no conjugate momentum and cannot participate in leapfrog MD.

2. B **is constrained to be flat**: the cocycle consistency requires the "field strength" of B to vanish. Just as the ordinary lattice curl dA sums a 1-form A_μ around the 4 edges of a plaquette, the curl dB sums the 2-form $B_{\mu\nu}$ over the 6 faces of a 3D cube. For a cube in the (μ, ν, ρ) directions at site x:

$$(dB)_{\mu\nu\rho}(x) = B_{\nu\rho}(x+\hat{\mu}) - B_{\nu\rho}(x) - B_{\mu\rho}(x+\hat{\nu}) + B_{\mu\rho}(x) + B_{\mu\nu}(x+\hat{\rho}) - B_{\mu\nu}(x) \quad (34)$$

Flatness requires this to vanish on every cube:

$$(dB)_{\mu\nu\rho}(x) = 0 \pmod N \qquad \forall\, x,\ \mu, \nu, \rho \quad (35)$$

Where does this come from? The flatness of B can be understood from two equivalent perspectives that correspond to the two origins of B itself:

From the cocycle perspective: The transition functions $\Omega_\mu \in SU(N)$ are ordinary matrices, and matrix multiplication is associative. The cocycle condition at 2D corners defines $B_{\mu\nu}$. At a 3D corner (where three directions μ, ν, ρ meet), the three possible orderings of $\Omega_\mu \Omega_\nu \Omega_\rho$ must all be consistent. Working out the $\mathbb{Z}_N$ phases, associativity forces $dB = 0 \pmod N$.

From the gauging perspective (Section 15.1): B was introduced by gauging the $\mathbb{Z}_N$ 1-form symmetry. The key point is that $\mathbb{Z}_N$ is a *discrete* group — it has no Lie algebra, no infinitesimal generators, no continuous curvature. Compare with gauging a continuous symmetry like $U(1)$: there the gauge field A_μ can have curvature $F = dA \neq 0$, which describes a propagating photon. For a discrete group, there is no analog — a discrete gauge field can only be flat. Physically, $dB \neq 0$ on a cube would be a $\mathbb{Z}_N$ topological defect (vortex). The $SU(N)/\mathbb{Z}_N$ theory only changes which bundles are allowed (boundary conditions), not which defects exist — excluding these vortices is exactly the flatness constraint.

These two perspectives are equivalent: gauging the 1-form symmetry and summing over all 't Hooft flux sectors are the same operation, and both require $dB = 0$.

The hierarchy mirrors electromagnetism:

	Electromagnetism	$\mathbb{Z}_N$ 2-form
Gauge field	A_μ on links	$B_{\mu\nu}$ on plaquettes
Field strength	$F_{\mu\nu} = dA$ on plaquettes	$(dB)_{\mu\nu\rho}$ on cubes
"Flat" means	$F = 0$ (no magnetic field)	$dB = 0$ (no B-curvature)
Consequence	A locally pure gauge	B locally pure gauge

So B is a **flat connection** with no local propagating degrees of freedom. It carries only global (topological) information.

In the path integral, B is not integrated over continuously but **summed** over:

$$Z = \sum_{B_{\mu\nu} \in \mathbb{Z}_N} \int [dU]\ e^{-S[U,B]} \quad (36)$$

In HMC, this sum is sampled by randomly proposing new B values and accepting/rejecting via Metropolis — fundamentally different from the leapfrog evolution of U.

$SU(N)$ **vs.** $SU(N)/\mathbb{Z}_N$

$$Z_{SU(N)/\mathbb{Z}_N} = \frac{1}{N^6} \sum_{B_{\mu\nu} \in \mathbb{Z}_N} Z_{SU(N)}(B_{\mu\nu}) \tag{37}$$

Local observables (including the topological susceptibility χ_t) agree in the large-volume limit: non-trivial B sectors contribute corrections suppressed as $e^{-\Delta F \cdot V}$.

13.4 Halfway-Updating HMC for Dynamic B

The Problem

$B_{\mu\nu} \in \{0, \ldots, N-1\}$ is discrete with no conjugate momentum — standard leapfrog cannot evolve it.

Algorithm

Place the B update at the midpoint of the MD trajectory:

1. Sample momentum $\pi \sim e^{-\pi^2/2}$
2. **First half-MD**: evolve $\{U, \pi\}$ with $H = \frac{1}{2}\pi^2 + S(U, B)$ for time $\tau/2$
3. **Random B update**: draw $B'_{\mu\nu}$ uniformly from $\{0, \ldots, N-1\}$ for each of 6 planes (symmetric proposal)
4. **Second half-MD**: evolve with $H(\cdot, B')$ for time $\tau/2$
5. **Metropolis**: accept $\{U', B'\}$ with $P_A = \min[1, e^{-\Delta H}]$

Detailed balance holds because forward and reverse paths are mirror images (leapfrog reversibility + symmetric B proposal).

Practical parameters: $\tau = 1.0$, $\Delta\tau = 0.02$, 25 leapfrog steps per half (50 total); acceptance $\sim 86\%$ at $\beta = 2.6$, $L = 20$.

Why It Resolves Topological Freezing

Since $Q_{\text{frac}} = -\varepsilon_{\mu\nu\rho\sigma} B_{\mu\nu} B_{\rho\sigma}/(8N)$, randomly resetting B each trajectory directly shuffles the topological sector — no gauge-field tunneling is required.

13.5 Topological Charge Measurement with B Field

Two Independent Topological Labels

	Bundle label $B_{\mu\nu}$	Instanton number n
Nature	6 discrete integers $\in \mathbb{Z}_N$	Continuous field tunneling
Barrier to change	None (flip an integer)	$\propto 1/a^2$, diverges
At fine a	Trivially easy	Catastrophically slow

Gradient Flow

The flow equation uses $S[V_t, B]$ (not $S[V_t]$): it drives $e^{-2\pi i B/N} P \to \mathbf{1}$, preserving the twist. Flow time: $t = (0.7L)^2/8$. After flow, ordinary plaquettes $\to \mathbf{1}$ while the corner plaquette $\to e^{+2\pi i B/N}$ — the fractional instanton structure is preserved.

Improved Definition

$$Q_{\text{imp}} = c_0\, Q_{\text{plaq}} + c_1\, Q_{\text{rect}}\,, \qquad c_0 = \tfrac{5}{3}, \quad c_1 = -\tfrac{1}{12} \tag{38}$$

The rectangle-improved definition reduces $O(a^2)$ discretization errors and converges best to half-integer values under gradient flow.

Fractional Q Formula

$$Q = -\frac{1}{N}\,\frac{\varepsilon_{\mu\nu\rho\sigma}\, B_{\mu\nu} B_{\rho\sigma}}{8} + \mathbb{Z} \tag{39}$$

The fractional part is entirely fixed by $B_{\mu\nu}$. For $SU(2)/\mathbb{Z}_2$ with $B_{12} = B_{34} = 1$: $Q = \frac{1}{2} + \mathbb{Z}$.

In algebraic topology language: the fractional part equals $-\frac{1}{N}\mathcal{P}_2(B) \pmod 1$, where $\mathcal{P}_2$ is the Pontryagin square of the $\mathbb{Z}_N$ 2-form B.

13.6 Numerical Results: $SU(2)$ vs. $SU(2)/\mathbb{Z}_2$

Table 3: Comparison of standard $SU(2)$ and $SU(2)/\mathbb{Z}_2$ with dynamic B field

Observable	$SU(2)$ (no B)	$SU(2)/\mathbb{Z}_2$ (with B)
Q history	Frozen near 0, -1; freeze ~ 500 MD	Freely jumps over all half-integers
$\tau_{\text{int}}(Q)$	$O(100)$ MD time	Essentially decorrelated
$\tau_{\text{int}}(E)$	$O(100)$ MD time	$O(20)$ MD time
τ_{int} scaling	$\propto a^{-2}$ (diverges)	Nearly flat $O(10)$ for all a
$\chi_t^{1/4}/\sqrt{\sigma}$	0.486(10)	Consistent (large-V limit)

The dynamic B field eliminates topological freezing: the autocorrelation time for Q drops from $O(100)$ to $O(1)$ MD time units, and its scaling with lattice spacing changes from $\propto a^{-2}$ (divergent) to approximately flat.

References for Section 13

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